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From Roster to Returns: An Integrated Model for Competitive-Commercial Strategy Optimization

Summary

Professional team management requires balancing short-term competitive gains with long-term financial sustainability—a coupled decision problem often addressed through subjective strategy. We develop a **Competitive-Commercial Coupled Framework** that integrates player capability, injury risk, win prediction, and franchise valuation into a prescriptive pipline. The pipline system enables general managers to evaluate roster strategies quantitatively under realistic constraints and uncertainty.

In the first part of the module, we construct an interpretable player impact score using position-specific metrics. We categorize basketball players by their on-court positions and apply **Principal Component Analysis** to select the five most influential indicators for each position based on weight. We introduce a nonlinear minutes factor to account for the tendency of skilled players playing long minutes. Crucially, we explicitly model season-ending injury risk via a logistic function of age, workload, and positional role, converting it into a risk-adjustment discount that yields a **risk-adjusted player value**.

We then calculate the minutes-weighted sum of risk-adjusted player values. The team's total player value is compared against a league-average benchmark to predict win rate. The predicted win rate is mapped through a **logistic-Elo response curve**, providing a stable preseason forecast. We adhere to a strict out-of-time validation protocol—training on data from 2021 to 2023and testing on 2024—which mirrors real-world decision-making and guards against overfitting.

We subsequently develop a **Parsimonious Valuation Model** incorporating a **Market Fundamental Score**. Franchise value is estimated via a log-linear regression with rate, payroll, and Market Score as indicators. This parsimonious specification achieves an out-of-sample average R^2 of 0.6157 across 28 NBA teams for the 2024 season, demonstrating robust predictive power for scenario analysis.

It did not escape our notice that variations in weight assignments lead to differences in model fit across teams. We apply **K-means**, dividing teams into 4 different clusters, based on their distinct optimal weight distributions. The resulting weight patterns reveal that these four team types are driven by different factors within the Market Score.

Finally, we establish a transparent, data-driven decision-support system through **actionable strategy optimization** and **expansion guidance**. The system quantitatively evaluates the combined effects of competitive risks and financial impacts in new markets, empowering managers to weigh personnel choices against financial outcomes under uncertainty—ultimately optimizing both win counts and long-term franchise value.

Keywords: PCA Weighting , Logistic Win-rate Forecast , Log-linear Regression

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1 Introduction

1.1 Restatement of the Problem

The rapid commercialization of professional sports has turned team management into a coupled decision problem: resources invested in on-court competitiveness can improve winning, yet the same resources also constrain financial performance. Under a limited-budget environment, decision makers must understand and quantify how player quality and roster spending translate into next-season team success, and how that success further converts into next-season business outcomes. Therefore, we aim to build a Competitive–Commercial Coupled Framework that links player performance, team winning, and financial returns in a unified, predictive, and decision-oriented manner.

Specifically, the model is required to accomplish the following tasks:

Task 1:Quantify player value and forecast next-season competitive performance. Using player statistics from the 2021-2024 seasons, the model should construct an interpretable player capability function and aggregate player capability to a team-level strength index.

Task 2:Infer team operating revenue from predicted performance. Establish a statistical relationship between competitive outcomes and operating revenue, and use this relationship to estimate next-season revenue under different performance scenarios.

Task 3:Predict team valuation from costs and financial structure. Incorporate cost drivers and financial statements to model and forecast team valuation, and quantify how financial structure mediates the translation from revenue/profit to valuation.

Task 4:Provide a data-driven strategy and optimization framework under realistic constraints. Formulate an actionable decision model that recommends optimal choices and respecting practical constraints such as budget caps, roster limits, risk tolerance, and scenario uncertainty. The framework should enable comparison of alternative strategies and identify robust, high-value solutions.

1.2 Our Work

To meet the requirements of the problem, this paper constructs a multi-module on-field performance–business coupling framework, organized along the main decision-making pipeline of "player capability → next-season win probability → next-season business outcomes → strategy optimization under constraints."

- First, at the player level, we develop a capability evaluation module that maps individual technical metrics, injury risk, and positional synergy effects into a unified capability representation suitable for season simulation.
- Second, at the competitive level, a win-rate prediction module transforms player capability into the expected next-season win percentage, thereby providing a quantitative characterization of team performance.
- Third, at the business level, we build a team valuation function driven primarily by win percentage and player expenses.

- Finally, at the strategy level, we explicitly incorporate constraints such as the salary cap, roster size limits, contract duration, and risk preferences, formulating an optimization problem that aims to maximize the expected overall return and minimize risk-related losses

Through these coupled modules, we build an end-to-end, computable pipeline that maps player-level capability to team win-rate forecast and further to next-season valuation, enabling constraint-aware strategy optimization and sensitivity analysis.

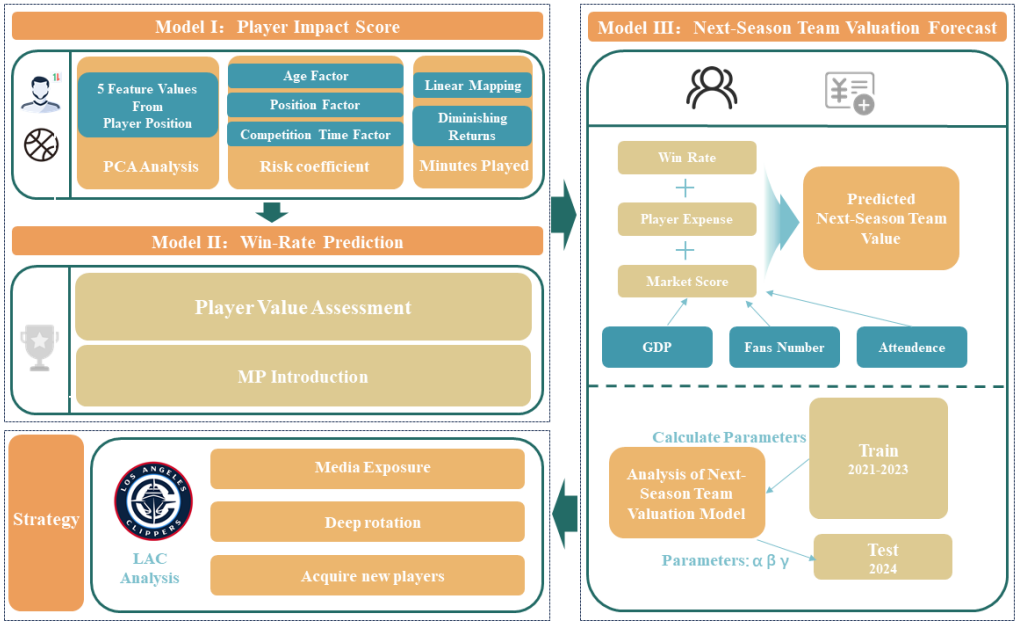

Figure 1: Workflow of the paper

2 Assumptions and Notations

The following assumptions are made:

- Over the forecasting horizon of one season, we assume league-wide rules and macroeconomic conditions do not experience regime shifts large enough to overturn learned relationships.
- We assume a player's latent capability estimated from individual performance indicators remains comparable from one season to the next.
- Because the exact future schedule and opponent sequence are not known when forecasting, we assume each team faces an "average league opponent" and ignore game-order effects.
- We assume roster moves can be abstracted into a finite candidate set and optimized under explicit constraints.Transaction frictions like negotiation failure, no-trade clauses are not modeled directly.

Table 1: Main Parameters and Their Descriptions

Symbol	Descriptions
i	Player index
k	Team index
t	Season index
p_i	Position of player i (PG/SG/SF/PF/C)
MP_i	Minutes played by player i in a season
Age_i	Age of player i
$Risk_i$	Estimated injury-risk probability for player i
V_i^{risk}	Risk-discounted player value score
S_k	Team strength score for team k
$\widehat{WR}_k$	Predicted win rate of team k for the next season
$TV_{k,t}$	Franchise valuation of team k in season t
$PE_{k,t}$	Player expenses of team k in season t

3 Data Preprocessing

Player-level indicators

We select player indicators according to two principles:

1. Game-impact relevance. We include efficiency (efg), shot volume proxies (FG, FGA), discipline/possession control (Tov, PF), and two-way impact summaries (ORtg, DRtg), together with availability (G, MP) and aging effects (Age).

2. Position discriminability. The Pos label is retained so that we can compare ability distributions across positions and perform position-aware imputation to avoid biasing low-minute players.

Because basketball statistics are often strongly correlated, we explicitly examine the correlation structure among candidate variables and use PCA later to form a compact ability representation.

Team-level indicators

Team features are chosen to support downstream tasks:

- Revenue generation and profitability: Revenue, Gate Receipts, and Operating Income.
- Cost structure and financial risk: PLAYER EXPENSES and DEBT/VALUE.
- Valuation and its drivers: TEAM VALUE and its decomposition into revenue-sharing, market size, arena, and brand, which allows interpretation of whether valuation is primarily performance-driven or market-driven.

3.1 Data Cleaning and Quality Control

Handling missing values and structural zeros

We distinguish structural zeros (valid outcomes) from missing/invalid entries.In our dataset, zeros/missing values are rare but can disproportionately affect low-minute players. To avoid overstating the discipline of such players, we adopt the following conservative rule:

- For PF and TOV, replace missing/invalid entries with the median of the league-wide position-specific distribution.

This reduces the chance that a low-minute player is incorrectly treated as having unrealistically low fouls/turnovers.

- For the remaining variables, we replace with 0.

4 Model 1:Player Impact Score

4.1 Indicator Weight Selection (PCA-based)

To avoid subjective tuning, we determine indicator weights in a fully data-driven way for each position group (PG/SG/SF/PF/C). Our procedure has two stages: making indicators comparable and letting PCA decide which indicators carry most information.

For a given position, we first transform raw indicators so they are comparable and numerically stable:

1. Percentage-type metrics (eFG) use an arcsine–square-root transform;
2. Count-type metrics (FG, FGA, TOV, PF, TRB, G) use $\log(1 + x)$;
3. Defensive rating DRtg is reversed so that "larger is better," and we further amplify DRtg differences for C and PG to reflect role-specific defensive importance;
4. Negative indicators (TOV, PF) are sign-flipped;
5. All retained numeric indicators are finally standardized by Z-score within the position group.

After standardization, we fit PCA on the processed indicators for that position and choose the smallest number of principal components whose cumulative explained variance exceeds 85%.

We then convert PCA results into indicator weights by accumulating each indicator's contribution across the selected components: indicators with larger absolute loadings on components that explain more variance receive higher weight. Concretely, we compute

$$w_k \propto \sum_{i=1}^{n_{\text{comp}}} \left(\text{explained_var}_i\right) \cdot |\text{loading}_{i,k}|$$

and normalize weights to sum to 1.

The table below is the result.

Table 2: Position-specific weight maps for efficiency score calculation (weights shown as decimals).

Position	Indicator 1	Indicator 2	Indicator 3	Indicator 4	Indicator 5
PG	FGA (0.2064)	G (0.2003)	PF (0.1972)	DRtg (0.1942)	ORtg (0.1896)
SG	PF (0.2186)	eFG (0.2155)	FGA (0.2060)	TOV (0.1981)	ORtg (0.1918)
SF	eFG (0.2090)	ORtg (0.2044)	FGA (0.1970)	FG (0.1926)	DRtg (0.1911)
PF	TOV (0.2117)	FGA (0.2086)	eFG (0.1948)	G (0.1902)	ORtg (0.1887)
C	eFG (0.2062)	FGA (0.2047)	PF (0.2000)	FG (0.1984)	G (0.1953)

4.2 Position-aware PCA Proxy

To quantify on-court skill while reducing redundancy among raw statistics, we construct a position-aware efficiency score inspired by PCA. Since different positions contribute through distinct roles, we compute the score using a position-specific set of key indicators and corresponding weights.

Let player i have position $p_i \in \{\text{PG, SG, SF, PF, C}\}$. Define a position-specific feature set $\mathcal{F}(p_i)$ (five indicators) and normalized weights $\{w_{p_i,k}\}_{k\in\mathcal{F}(p_i)}$ such that $\sum_k w_{p_i,k} = 1$. All input indicators are standardized prior to scoring, denoted by $x_{i,k}$. The efficiency score is

$$E_i = \sum_{k\in\mathcal{F}(p_i)} w_{p_i,k} x_{i,k}. \tag{4.2}$$

In implementation, the weights are fixed(Table 2) and reflect the relative contribution of each indicator within the position group.

4.2.1 Minutes-Played Factor with Diminishing Returns

Pure efficiency scores may overrate players with limited playing time, because small-sample performance can be noisy and because availability is essential for season-level contribution. We therefore introduce a minutes-played factor that increases with playing time but exhibits diminishing returns.

Let $MP_i^{(\cdot)}$ denote a minutes measure used for valuation.In this work, the minutes measure used for valuation is the rank-calibrated normalized minutes MP_i^{norm} defined in Section 4.4. We define a reasonable maximum season workload as $MP_{\max} = 82 \times 35 = 2870$,

and compute the capped minutes ratio

$$r_i = \min\left(1, \frac{MP_i^{(\cdot)}}{MP_{\max}}\right).$$

The minutes factor is

$$f_{MP}(i) = 0.2 + 0.8\sqrt{r_i}.$$

This design guarantees $f_{MP}(i) \in [0.2, 1]$: even low-minute players retain a baseline contribution (0.2), while the square-root shape prevents extreme dominance by workload alone and reflects diminishing marginal returns of additional minutes.

4.2.2 Rank-Calibrated Minute Normalization

Skilled players tend to play longer minutes on court. By data analysis, we discovered players' time in matches have significant linear correlation with players' rank. Therefore, we can estimate an expected minutes value from a player's rank Rk_i via a linear mapping(R^2=0.907):

$$\hat{MP}_i = 1485.9 - 3.24Rk_i. \tag{4.2.2a}$$

We then compare the observed minutes MP_i to this baseline via a log-ratio adjustment, which is less sensitive to extremes than raw ratios. Define

$$\phi_i = \text{clip}\left(1 + \ln\left(\frac{MP_i}{\hat{MP}_i}\right),\ 0.5,\ 2.0\right), \tag{4.2.2b}$$

and the normalized minutes are

$$MP_i^{\text{norm}} = \hat{MP}_i \cdot \phi_i, \qquad (\text{if } \hat{MP}_i > 0;\ \text{otherwise } MP_i^{\text{norm}} = MP_i). \tag{4.2.2c}$$

The clipping bounds $[0.5, 2.0]$ prevent extreme deviations from dominating subsequent valuation, while the log adjustment provides a smooth correction when observed minutes differ moderately from the rank-based expectation.

4.3 Injury Risk and Risk-Discounted Player Value

Availability risk is crucial for forward-looking win-rate prediction, since a player's contribution depends not only on skill and workload but also on the probability of missing time. We model injury risk using a logistic function combining age, workload, and position effects.

Let Age_i be player age and MP_i be observed minutes. We define an age baseline at 27 years and use a piecewise age slope to reflect faster risk growth for veteran players:

$$AgeEffect_i = (Age_i - 27) \times \begin{cases} 0.03, & Age_i \le 27, \\ 0.09, & Age_i > 27. \end{cases} \tag{4.3a}$$

Workload effect is modeled by a log transform (to capture saturation):

$$LoadEffect_i = 0.6 \cdot \ln\left(\frac{MP_i}{677} + 1\right), \tag{4.3b}$$

and we add a position penalty for heavier-contact frontcourt roles:

$$PosPenalty_i = \begin{cases} 0.2, & p_i \in \{C, PF\}, \\ 0, & \text{otherwise.} \end{cases} \tag{4.3c}$$

The latent risk score and injury risk are

$$z_i = -2.0 + AgeEffect_i + LoadEffect_i + PosPenalty_i,$$
$$Risk_i = \frac{1}{1 + \exp(-z_i)}. \tag{4.3d}$$

Figure 2 provides an interpretable visualization of our risk mechanism.We summarize $Risk_i$ by age group, and the distributions exhibit a clear age-gradient: younger groups concentrate at lower risk levels, while older groups shift to higher central tendency and larger spread.

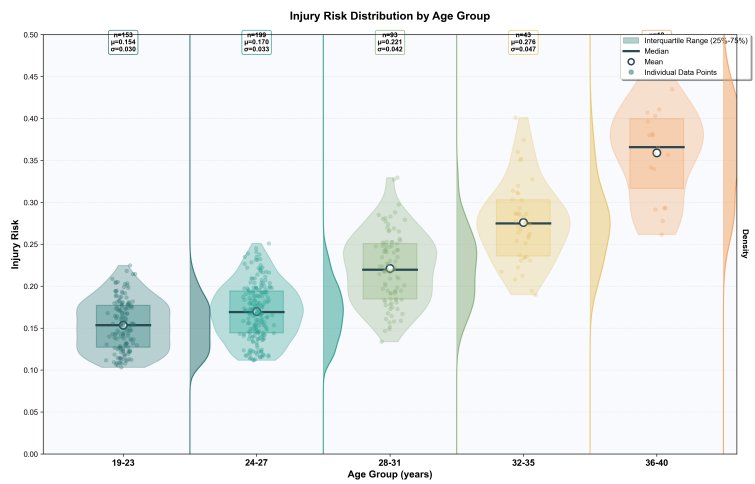

Figure 2: Injury Risk Distribution by Age Group

Two patterns align with basketball domain intuition and support forward-looking use: aging and heavy minutes jointly elevate risk, indicating that "high-usage veterans" are the most fragile cohort; and the post-27 acceleration is consistent with the model design that veterans experience faster risk growth.

Finally, we apply a risk discount to the value contribution. Let λ denote the risk impact coefficient ($\lambda = 0.2$). The risk discount is

$$D_i = 1 - \lambda Risk_i, \tag{4.5e}$$

and the risk-adjusted player value becomes

$$V_i^{\text{risk}} = (E_i \cdot f_{MP}(i)) \cdot D_i. \tag{4.5f}$$

This formulation preserves interpretability: E_i captures position-aware efficiency, $f_{MP}(i)$ captures workload with diminishing returns, and D_i discounts expected contribution under availability risk.

5 Model 2:Win-Rate Prediction via Team Strength and Elo

To connect individual player capability with team-level competitive outcomes, we develop a win-rate prediction module that aggregates player scores into a single team strength index and then maps the strength gap into game-winning probabilities using an Elo/Logistic formulation.

Let i index players and k index teams. From Model 1, each player i has a normalized capability score V_i . We also use the previous season's minutes per game MPG_i to approximate the player's expected on-court participation next season. Team rosters are denoted by R_k. The Elo scaling constant is fixed at $D = 400$.

5.1 Weighted Team Strength Construction

A key design choice is how to transfer a roster into a single team strength number. Since basketball outcomes are dominated by who is actually on the court, we weight each player's capability by their expected playing time share. For team k, define the minutes-share weight:

$$w_{i,k} = \frac{MPG_i}{\sum_{j \in R_k} MPG_j}, \quad i \in R_k.$$

This guarantees $\sum_{i \in R_k} w_{i,k} = 1$, making the team score comparable across teams and seasons.

The minutes-weighted team strength is then:

$$S_k = \sum_{i \in R_k} w_{i,k} V_i.$$

S_k represents the expected team strength under a stable rotation assumption. This is intentionally parsimonious: it avoids overfitting while still embedding the key causal channel—better players playing more minutes raises team strength.

To reflect roster fit and depth, we define a positional synergy coefficient $\lambda_{\text{syn}} \in [0, 1]$ from the full-roster position counts over $\{C, PG, SG, SF, PF\}$. Let $\mathbf{c}_k$ be the team's position-count vector and $\mathbf{c} = [2, 3, 3, 4, 3]$ the ideal mix; we measure imbalance by $d_k := \|\mathbf{c}_k - \mathbf{c}\|_2$ and normalize it by a worst-case distance $d_k^{\max}$ (all players concentrated at one position), giving

$$\lambda_{\text{syn}} = \max\left(0, 1 - \frac{d_k}{d_k^{\max}}\right).$$

We then apply a small multiplicative uplift to the core-rotation value: so balanced rosters receive a modest bonus while highly imbalanced rosters do not.

5.2 From Team Strength to Next-Season Team Win Rate

After obtaining each team's composite strength score S_k from minutes-weighted player capability, we convert team strength into a next-season win-rate forecast using an Elo/Logistic mapping. Since the probability of winning is determined by the strength gap between a given team and a reference opponent, and the Elo logistic curve provides a stable, interpretable transformation from "gap" to "probability."

Because the exact next-season schedule may be unknown at the preseason decision stage, we adopt a schedule-agnostic but informative approximation: we forecast each team's win probability against a league-average opponent. In our implementation, the computed league average is $S_{\text{league}} = 969.33$.

Given a team with strength S_k, its neutral-court probability of winning against the league-average opponent is defined by the Elo formula with $D = 400$:

$$p_k = \mathbb{P}(\text{Win} \mid k) = \frac{1}{1 + 10^{-\frac{S_k - S_{\text{league}}}{400}}}.$$

This choice preserves the standard Elo interpretation: as a team's strength exceeds the league mean, its win probability rises smoothly above 50%; conversely, teams below the mean fall below 50%.

Under the league-average opponent approximation, we take the next-season win rate as the expected win probability: $\hat{WR}_k = p_k$.

This yields a preseason, schedule-agnostic forecast that is especially suitable for business modeling and strategy optimization, because it provides a single, comparable performance metric for every team.

Applying the above mapping to all 30 teams produces a full predicted win-rate table. The results are centered around $\hat{WR} = 0.500$, with a highest predicted win rate of 0.627 (SAS) and a lowest of 0.341 (DAL). The top of the ranking includes SAS (0.627), NYK (0.625), and DET (0.610), reflecting their above-average team strength values under our minutes-weighted aggregation and Elo transformation, which approach the current NBA standings.

5.3 Evaluation Metrics for Forward-Looking Validation

To evaluate predictive accuracy, we use a strictly forward-looking protocol: parameters and transformations are fitted on seasons up to year T, and we forecast win rates for season $T + 1$. Let WR_k denote the realized next-season win rate and $\hat{WR}_k$ our prediction. We report:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}\left(WR_k - \hat{WR}_k\right)^2}, \quad R^2 = 1 - \frac{\sum_k \left(WR_k - \hat{WR}_k\right)^2}{\sum_k \left(WR_k - \hat{WR}\right)^2}.$$

RMSE penalizes large misses, and R^2 measures improvement over a baseline that predicts the average win rate for all teams.

Although the Elo provides a stable preseason forecast, some teams exhibit persistent team-specific bias. We apply an **outlier-factor correction** that learns each team's historical deviation tendency under a strict time-alignment setting: we use the prediction produced from season t information to explain the realized win rate in season $t+1$. Let the one-step-ahead residual be $e_{k,t+1} = WR_{k,t+1} - \overline{WR}_{k,t+1}$.

We estimate a team-specific average deviation $\bar{e}_k$ on the training window and convert its magnitude into a bounded base factor $b_k = \min(2|\bar{e}_k|, 1)$. Next, we incorporate directional consistency: if the residual signs over the most recent three seasons are aligned in at least two seasons, we add a small bonus Δ_k to reflect that the bias is likely structural rather than noise. The corrected forecast is then obtained by adjusting the original prediction toward the historically observed direction,

$$\overline{\mathrm{WR}}^{\mathrm{corr}}_{k,t+1} = \mathrm{clip}\left(\overline{\mathrm{WR}}_{k,t+1} + (b_k + \Delta_k) \cdot \bar{e}_k,\ 0,\ 1\right),$$

As shown in Figure 3, the corrected model produces a noticeably tighter error distribution across teams, indicating that the correction effectively removes persistent team-level bias while preserving the parsimonious structure.

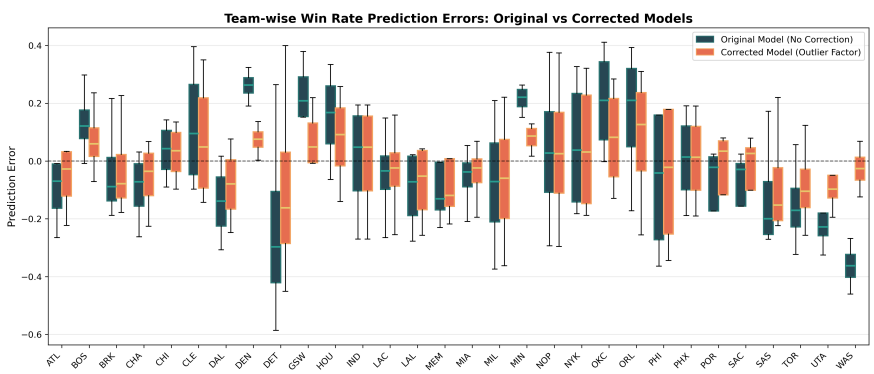

Figure 3: Team wise Win Rate Original vs Corrected Models

6 Model 3:Next-Season Team Valuation Forecast

Model 3 forecasts next-season franchise valuation using a small set of drivers that are observable at the preseason decision stage and interpretable for strategy design. A Pearson-correlation screening indicates that Team Value is positively associated with competitive performance, player expenses, and market/operation proxies such as attendance, social-following measures, and local GDP. To balance interpretability and generalization under limited seasons, we incorporate these signals through a parsimonious log-linear valuation model with an explicit market factor.

Because franchise value is strictly positive and evolves multiplicatively, we model valuation on the log scale. This improves numerical stability and allows coefficients to be read as approximate percentage effects, which is useful for translating decisions into business implications.

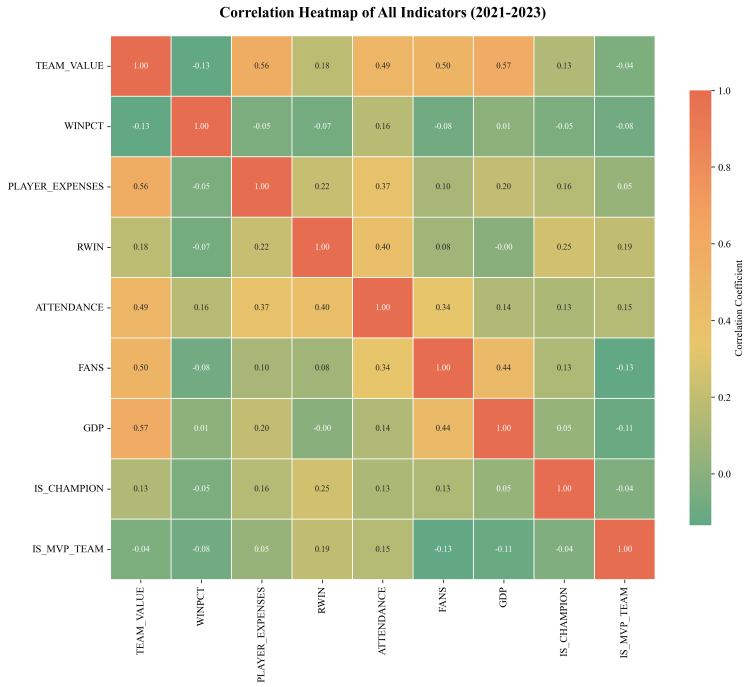

Figure 4: Correlation Heatmap

6.1 Log-Linear Valuation Model

Notation. Let $TV_{k,t}$ denote franchise valuation for team k in season t. Let $WR_{k,t}$ be the realized regular-season win rate and $PE_{k,t}$ be player expenses. To capture persistent, hard-to-measure franchise advantages (market size, brand premium, arena economics) while retaining portability to expansion scenarios, we construct a market score MS_k from observable market indicators.

Market score construction. For each team k, we use the most recent pre-forecast market indicators and normalize them to comparable scales:

$$GDP_k^{(n)} = \frac{GDP_k}{1000}, \qquad ATT_k^{(n)} = \frac{ATT_k}{100000}, \qquad FANS_k^{(n)} = FANS_k.$$

Based on the Pearson correlation screening (Figure 4), we retain three market proxies—local GDP, social following (FANS), and attendance (ATT)—as the dominant non-performance drivers of Team Value. To avoid arbitrary weighting, we use a two-stage procedure. First, the Pearson correlations provide a prior importance ordering and restrict candidates to these three variables. Second, we perform a grid search over nonnegative weights $(w_{GDP}, w_{FANS}, w_{ATT})$ under the simplex constraint $w_{GDP} + w_{FANS} + w_{ATT} = 1$, and select the combination that maximizes the average out-of-sample fit of the valuation regression on the validation protocol.

The best-performing weights are:

$$w_{\text{GDP}} = 0.2400, w_{\text{FANS}} = 0.2200, w_{\text{ATT}} = 0.5400$$

yielding a validation $R^2 = 0.6157$ for the market-score component.

We therefore define

$$MS_k = 0.2400 \cdot GDP_k^{(n)} + 0.2200 \cdot FANS_k^{(n)} + 0.5400 \cdot ATT_k^{(n)},$$

For valuation equation, the final model is a pooled log-linear regression:

$$\ln(TV_{k,t}) = \alpha + \beta\, WR_{k,t} + \theta\, PE_{k,t} + \gamma\, MS_k + \varepsilon_{k,t}.$$

Compared with team-by-team fitting, this pooled specification reduces variance under short time series and, importantly, enables valuation prediction for new or relocated franchises by assigning them a plausible MS_k from local market indicators.

Estimation and forecasting. Parameters $(\alpha, \beta, \theta, \gamma)$ are estimated by ordinary least squares on seasons 2021–2023. For next-season forecasting, we plug in $\widehat{WR}_{k,t+1}$ from Model 2, the decision-controlled payroll $PE_{k,t+1}$, and the market score MS_k. The raw prediction is

$$\widehat{TV}_{k,t+1}^{raw} = \exp\Big(\alpha + \beta\, \widehat{WR}_{k,t+1} + \theta\, PE_{k,t+1} + \gamma\, MS_k\Big).$$

In validation, the raw model exhibits a mild systematic underestimation. We therefore apply a single multiplicative calibration factor κ (estimated from the validation bias) to correct the scale:

$$\widehat{TV}_{k,t+1} = \kappa \cdot \widehat{TV}_{k,t+1}^{raw}, \qquad \kappa = 1.084.$$

6.2 Model Test and Analysis

We adopt a strictly forward-looking test: train on 2021–2023 and validate on 2024. We report R^2, MAPE, RMSE, and MAE on the held-out 2024 valuations.

Estimated coefficients. Training on 2021–2023 yields:

$$\ln(TV) = 6.385 + 0.036 \cdot WR + 0.0079 \cdot PE + 0.0657 \cdot MS.$$

The training R^2 is 0.6203, indicating that a compact set of fundamentals already explains a majority of valuation variation across teams and seasons.

Out-of-sample performance (2024). After applying calibration $\kappa = 1.084$, the 2024 validation achieves MAPE = 15.16%. In terms of practical accuracy, 27/28 teams fall within this 95% interval, indicating that the vast majority of forecasts deviate from realized valuations by an amount consistent with the model's noise level; the remaining outlier is consistent with exceptional brand/narrative premiums beyond the three-fundamental specification.

Interpretation of drivers. A standardized contribution check shows that payroll and market structure dominate the predictive signal (player expenses contribution ≈ 0.232, market score contribution ≈ 0.213), while the marginal effect of one-season win rate is comparatively small in our dataset. This aligns with sports-business intuition: short-horizon valuation largely reflects

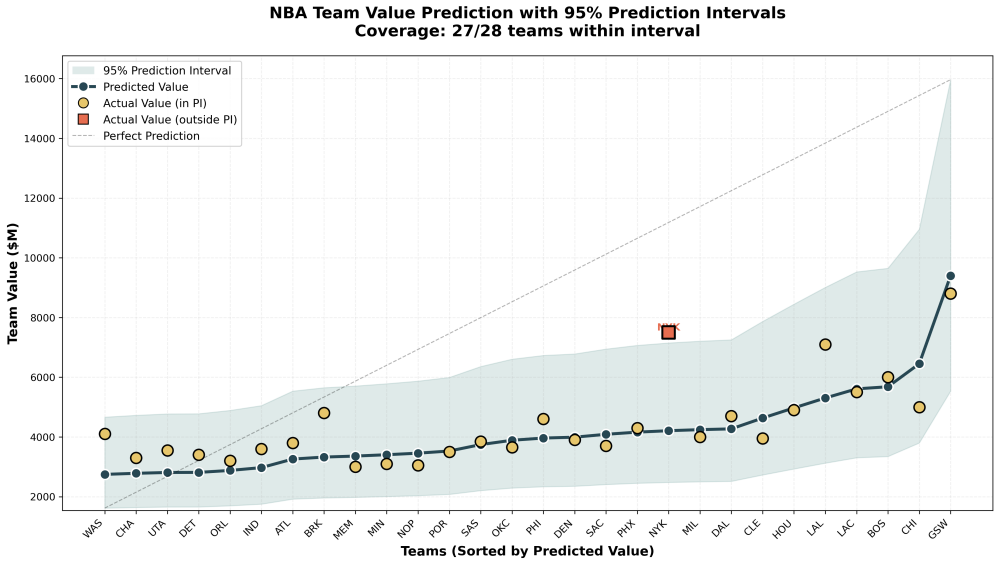

Figure 5: NBA Team Value Prediction

persistent market demand and monetization capacity, while competitive outcomes matter but are partially capitalized through spending and brand.

The largest errors concentrate on teams with exceptional brand or narrative premiums not fully captured by our three fundamentals. For example, NYK is underpredicted and MIA is overpredicted in 2024. This suggests that "star shocks" can create valuation deviations beyond fundamentals. In our framework, these effects are handled downstream in the decision module via a separate business influence term rather than by inflating the core valuation model with many ad hoc variables.

7 Decision and Evaluation Module

Building on Model 1–3, we translate analysis into action by constructing a decision-and-evaluation module that recommends feasible roster and operational moves under practical constraints. The key idea is to treat player acquisition as the controllable lever that changes the team's aggregated strength, shifts next-season win probability, and ultimately reshapes next-season business outcomes and franchise value. Therefore, every candidate decision can be scored consistently on a unified competition–business scale before we recommend an implementable strategy.

7.1 Roster Acquisition

We model player acquisition as a 0–1 integer programming (IP) problem and explicitly consider two acquisition channels: free agency and the draft. Each candidate player i is associated with a salary S_i, an on-court performance score PS_i, and a business influence score B_i that captures star effects. The binary decision variable $x_i \in \{0, 1\}$ indicates whether the player is signed. This formulation directly follows the document's IP setup and variable definitions.

We operationalize the objective as a team composite score that balances competitive gains driven by the total acquired performance PS, and business gains driven by total brand influence B, while recognizing payroll as the binding resource. Concretely, we maximize a weighted score of

on-court strength + off-court brand, and then evaluate the next-season consequences by passing the resulting roster into Model 2 and Model 3. This design ensures that roster choices are selected for their actionable upside, but ultimately judged by their forecasted next-season outcomes, consistent with the paper's coupling philosophy.

7.2 Realism Constraints

We incorporate three layers of realism constraints. First, a salary-cap constraint ensures that the total payroll of acquired players does not exceed the available budget: $\sum_i x_i S_i \leq C_{\text{cap}}$

Second, a roster-size constraint ensures the signed players fit the team's target headcount: $\sum_i x_i = 15$

Third, a positional-balance constraint prevents unbuildable rosters: based on the PCA-derived role taxonomy, we restrict each position group to lie within a practical interval to maintain lineup feasibility:

$$2 \leq \sum_{i \in \text{pos } p} x_i \leq 5, \quad \forall p$$

In addition, we enforce mandatory retention for existing core players with guaranteed contracts, making the optimization reflect the front-office reality that not all roster slots are freely adjustable.

8 League Expansion Scenarios

8.1 Injury-Risk Adjustment

When the new franchise is geographically remote, league-wide travel distance increases, amplifying fatigue accumulation and raising injury exposure. We quantify this shock through a travel fatigue factor built from the schedule-weighted travel burden. Let T denote the set of all teams, D_k the set of teams in the same division as team k, and C_k the set of teams in the same conference as team k. Let d_{kj} be the geographic distance between teams k and j, and let n_1, n_2, n_3 denote the number of games played against same-division opponents,same-conference but different-division opponents, and different-conference opponents, respectively. The incremental travel burden induced by expansion is modeled as

$$\Delta D_k = \sum_{j \in D_k \setminus \{k\}} d_{kj} n_1 + \sum_{j \in C_k \setminus D_k} d_{kj} n_2 + \sum_{j \in T \setminus C_k} d_{kj} n_3.$$

For cross-team comparability, we define the travel fatigue factor as the relative increase over a baseline travel load D_k^{base}:

$$\text{TF}_k = \frac{\Delta D_k}{D_k^{\text{base}}}.$$

Let TF_k represent the percentage increase in travel distance caused by the new location. We then adjust the injury-risk coefficient multiplicatively:

$$\text{Risk}_{\text{new}} = \text{Risk}_{\text{old}} \left(1 + \beta \cdot \text{TF}_k\right).$$

β controls how sensitively injury risk responds to travel burden.

Operationally, the document's implication is direct: if $Risk_{\text{new}}$ becomes too high, teams must reduce core players' minutes to prevent risks, this is implemented by lowering the minutes weights assigned to high-usage core players and reallocating minutes to bench players, which moderates the effective team strength used in Model 2 and generates a more realistic next-season win-rate forecast under high-travel regimes.

8.2 Revenue Sharing and Valuation Impact Under Expansion

Expansion produces a two-sided valuation effect. On the positive side, the new franchise generates an entry fee and—if located in a large market—may raise media-contract value; on the negative side, total league revenue must be shared among $N + 1$ teams instead of N. Let R_{total} denote pre-expansion league revenue, and let $\Delta R(\text{Loc}_{\text{new}})$ denote the incremental revenue premium contributed by the new location. The post-expansion per-team share is modeled as:

$$R_{\text{share}}(\text{Loc}_{\text{new}}) = \frac{R_{\text{total}} + \Delta R(\text{Loc}_{\text{new}})}{N + 1}.$$

We then define an impact metric by comparing the post-expansion share to the pre-expansion share:

$$\text{Impact} = \frac{R_{\text{total}} + \Delta R(\text{Loc}_{\text{new}})}{N + 1} - \frac{R_{\text{total}}}{N}.$$

If Impact > 0, expansion is financially beneficial to incumbents; otherwise, it is harmful.

Based on the model, we classify candidate locations into two extreme regimes that yield clear managerial guidance. **Beneficial locations** are characterized by **large market and geographic proximity**: $\Delta R(\text{Loc}_{\text{new}})$ is large because of higher media value, while ΔD_k is small or even negative, so risk does not rise materially. In this regime, teams should support expansion and treat additional revenue share as an opportunity to reinvest in performance and player services.

In contrast, **harmful locations** are **small market and geographically remote**: $\Delta R(\text{Loc}_{\text{new}})$ is too small to offset dilution, while ΔD_k rises sharply, increasing fatigue-driven risk.We recommend avoiding such expansion if possible; if unavoidable, franchises should pivot toward a long-term roster strategy ,such as acquiring younger, more durable players and developing through the draft, and apply travel-aware load management to protect core-player risk and preserve overall season performance.

8.3 Additional commercial issues and injury coping mechanisms

To make our management framework operational, we design a key-player injury contingency mechanism. We first categorize franchises using the multi-dimensional profile clustering illustrated in Figure 6, including big market(fan economy), small market, balenced market and market premium,

then select the Los Angeles Clippers (LAC) as a representative star-driven contender case for stress testing.

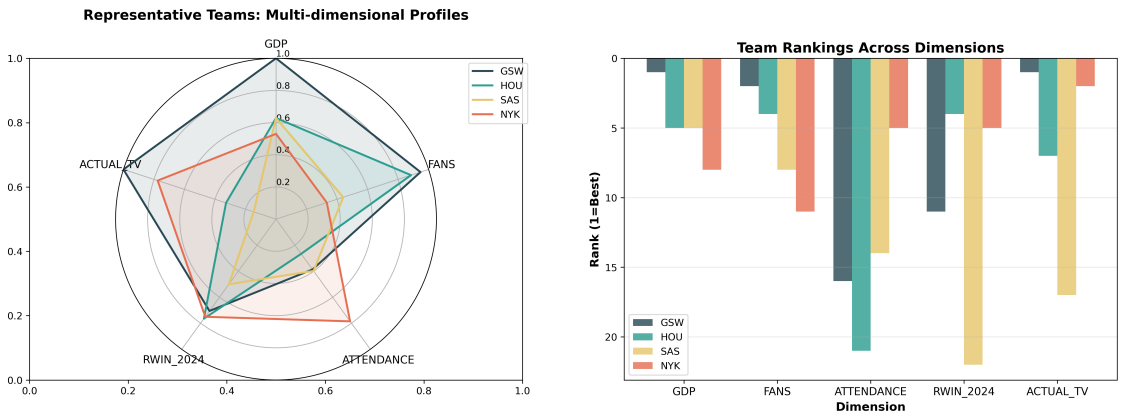

Figure 6: Representative Teams

Our competitive module defines a minutes-share weighted team strength, if the franchise player is injured, we set his minutes share to zero and re-normalize the remaining rotation. For LAC, the computed outputs in our code-based scenario show that Harden's injury reduces:(i)Team strength: 1487.70 → 1307.55. (ii)Predicted win rate: 0.636 → 0.476. (iii)Estimated team value: 32.39B → 32.13B. This reveals an important management insight: the competitive drop is large, while the valuation drop is comparatively modest.

9 Sensitivity Analysis

To ensure that Model 3 is reliable for decision-making (roster/finance/expansion), we conduct sensitivity tests from two complementary angles: **robustness of the market-score construction**, and **residual-based sensitivity** that checks whether the prediction error systematically depends on any key driver.

9.1 Robustness of market-score weights

Our market score compresses multiple market proxies into a single covariate, but the relative weights may influence performance. Therefore, we perform a grid search over nonnegative weights

$$(w_{\text{GDP}}, w_{\text{FANS}}, w_{\text{ATT}})$$

under the simplex constraint $w_{\text{GDP}} + w_{\text{FANS}} + w_{\text{ATT}} = 1$, and evaluate the 2024 validation fit across the feasible region (Figure 7).

The performance landscape exhibits a clear optimum around the selected weights, and the surrounding region remains relatively smooth, indicating that the chosen combination is not a "knife-edge" solution. Meanwhile, moving toward extreme corners (over-emphasizing one proxy while suppressing others) degrades validation performance quickly, implying that the market factor must jointly reflect **macro purchasing power (GDP), fan base (FANS), and revealed demand through attendance (ATT)** rather than relying on a single indicator.

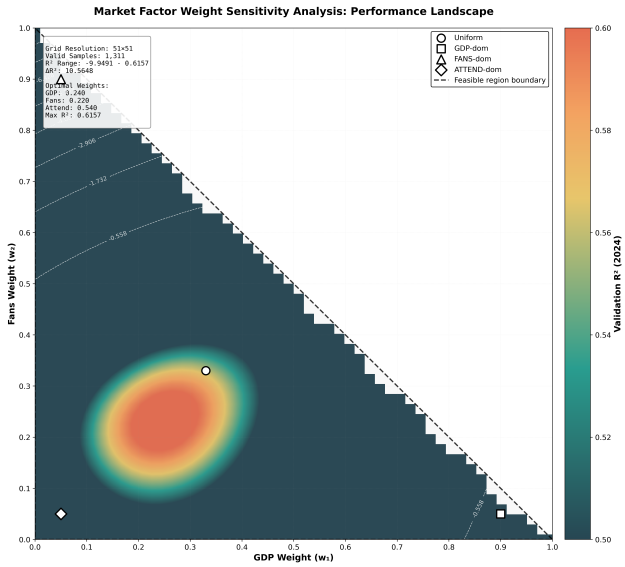

Figure 7: Market Factor Weight Sensitivity Analysis

9.2 Residual-based parameter sensitivity on the 2024 hold-out

We further test whether prediction errors reveal systematic bias across the range of each predictor.Using the 2024 validation set, we analyze three drivers by plotting team-level errors against each predictor and overlaying a binned mean trend with dispersion bars and a fitted linear trend (the figure included in this section). Pearson correlation tests show that none of the three predictors is significantly correlated with the error at the 5% level: $r_{\mathrm{WR}} = 0.2936$ ($p = 0.1295$), $r_{\mathrm{PE}} = 0.3502$ ($p = 0.0677$), and $r_{\mathrm{MS}} = 0.3233$ ($p = 0.0933$). This indicates that the model does not exhibit strong systematic over-/under-estimation that worsens monotonically with any single input, supporting stability across small/large-market teams and low/high payroll regimes. The mild positive tendencies (especially for payroll and market score) suggest that a small subset of high-profile teams may still carry additional "brand premium", consistent with the residual discussion shown in Figure 8.

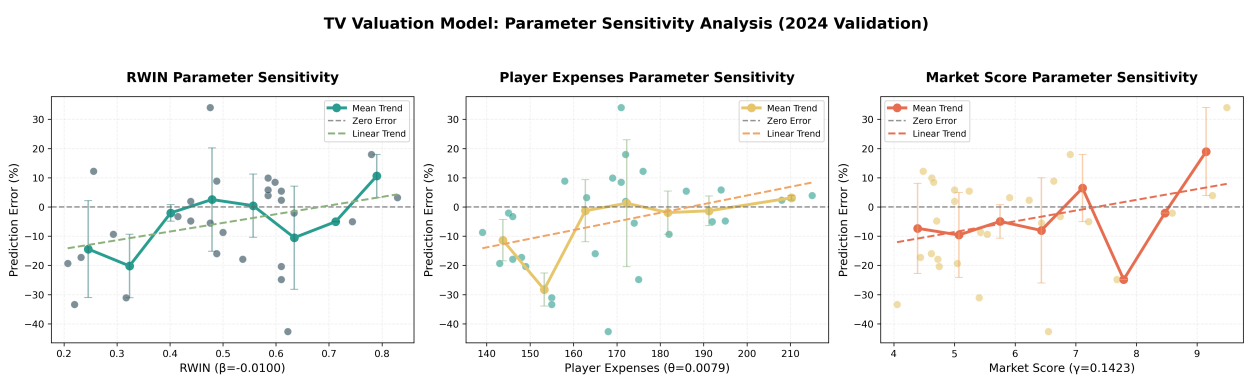

Figure 8: Parameter Sensitivity Analysis

9.3 Importance comparison and practical error concentration

A standardized contribution comparison $|\beta|\sigma(\text{RWIN})$, $|\theta|\sigma(\text{PE})$, $|\gamma|\sigma(\text{MS})$ shows that valuation is dominated by structural fundamentals: market score (57.2%) and player expenses (42.4%), while WR contributes only 0.4%. Overall, the mean absolute percentage error is 13.41%, with error standard deviation 16.81% (max positive 33.99%, max negative -42.67%). In distribution terms, 7/28 teams fall within ±5% error, 16/28 within ±10%, and 17/28 within ±15%. Therefore, Model 3 is sufficiently robust for ranking and scenario comparison, while the largest deviations motivate adding an optional "brand shock" adjustment term when analyzing elite global-market franchises.

10 Strengths and Weaknesses

Strengths

- **End-to-end coupled decision pipeline.** The paper builds a clear "player capability → win probability → business outcomes → strategy optimization" chain, so every modeling block directly serves the final decision objective rather than existing as an isolated predictor.
- **Interpretability with disciplined feature engineering.** The minutes factor uses a capped workload ratio and a square-root form to enforce diminishing returns, while rank-calibrated minute normalization reduces rotation noise and prevents extreme minutes from dominating downstream team strength.
- **Risk is integrated** Injury risk is explicitly modeled with a logistic form combining age, workload, and a position penalty, and then translated into a risk discount that directly modifies player value, making the uncertainty mechanism operational in later modules.
- **strong generalization capability** Next-Season Team Valuation Forecast Model is sensitive to market factors, giving the model high potential for optimization. The Winning Rate Prediction Model is built upon the fundamental characteristics of basketball as a sport and does not rely on specific datasets. Its strong generalization capability allows it to make predictions not only for the NBA, but also for other leagues such as the WNBA, CBA, and beyond.

Weaknesses

- **Schedule simplification may bias win forecasts.** The model assumes an average league opponent and uses a schedule-agnostic mapping, which can miss schedule strength, conference imbalance, and opponent-specific matchups—especially important for teams with extreme travel or divisional difficulty.
- **Valuation residuals reveal missing brand shock drivers.** The paper itself notes larger errors for teams with exceptional brand premiums not captured by the three-fundamental specification, meaning some high-profile cases can be systematically under-/over-predicted.
- **Transaction frictions are abstracted away.** The assumptions acknowledge that negotiation failure, no-trade clauses, etc. are not modeled; this can make the optimization output optimistic in feasibility when applied to real front-office conditions.

References

[1] Hvattum, L. M., & Arntzen, H. (2010). Using ELO ratings for match result prediction in association football. *International Journal of Forecasting*, 26(3), 460–470. https://doi.org/10.1016/j.ijforecast.2009.10.002

[2] Humphreys, B. R., & Mondello, M. (2008). Determinants of Franchise Values in North American Professional Sports Leagues: Evidence from a Hedonic Price Model. *International Journal of Sport Finance*, 3(2), 98–105. https://doi.org/10.1177/155862350800300202

[3] Payyappalli, V. M., & Zhuang, J. (2019). A data-driven integer programming model for soccer clubs' decision making on player transfers. *Environment Systems and Decisions*, 39(4), 466–481. https://doi.org/10.1007/s10669-019-09721-7

[4] Smith, Z. J., & Bickel, J. E. (2023). A roster construction decision tool for MLS expansion teams. *Journal of Quantitative Analysis in Sports*, 19(1), 1–14. https://doi.org/10.1515/jqas-2021-0041

[5] Muniz, M., & Flamand, T. (2023). Sports analytics for balanced team-building decisions. *Journal of the Operational Research Society*, 74(8), 1892–1909. https://doi.org/10.1080/01605682.2022.2118634

[6] Barnes, S., Bjarnadóttir, M., Smolyak, D., & Thiele, A. (2024). A data-driven optimization approach to baseball roster management. *Annals of Operations Research*, 335(1), 33–58. https://doi.org/10.1007/s10479-023-05725-4

[7] Jolliffe, I. T. (2002). *Principal Component Analysis*. Springer Series in Statistics. https://doi.org/10.1007/b98835

[8] Bertsimas, D., & Sim, M. (2004). The Price of Robustness. *Operations Research*, 52(1), 35–53. https://doi.org/10.1287/opre.1030.0065

[9] Statista. n.d. "The Statistics Portal." Accessed February 2, 2026. https://www.statista.com.

[10] Sports Reference LLC. n.d. "Basketball Reference." Accessed February 2, 2026. https://www.basketball-reference.com/.

Letter

To: Team Owner and General Manager
From: Team #2607299
Subject: An Integrated Competitive + Commercial Strategy for Our Team
Date: February 3, 2026

Dear Owner and General Manager,

The attached memo outlines a unified strategy built on our Competitive–Commercial Coupled Framework. This approach moves beyond isolated decisions—it links roster construction, win forecasting, and franchise valuation into a single coherent system. Our goal is to deliver a disciplined operating plan that protects competitive upside while ensuring long-term financial resilience.

The core of our proposed strategy is to establish a win-now posture with built-in resilience on the court, while cultivating a commercial foundation that is anchored in durable market fundamentals rather than transient win-loss cycles.

This is achieved through a risk-aware approach to roster construction—prioritizing minutes-weighted strength, ensuring at least two-deep redundancy for high-usage roles, and applying injury-adjusted valuations to all major commitments—alongside a commercial plan that actively invests in team identity, fan engagement and game-day experience as leading indicators of franchise health.

In managing the inherent trade-offs, we advocate aggressive moves only when near clear competitive thresholds; otherwise we preserve flexibility and develop youth, framing resource allocation as a portfolio that pursues wins without creating financial fragility.

To safeguard this path, we explicitly mitigate financial-leverage risk through conservative debt and structured contracts, address injury risk via a three-layer hedge of prevention, depth, and contingency planning, and treat expansion uncertainty as a scenario-based planning exercise to guide roster-age and investment decisions.

Our framework avoids the classic trap of chasing wins at the expense of value—or protecting profits while eroding competitive identity. It forces every major decision to be evaluated for both its competitive impact and its commercial effect, creating a virtuous cycle: a risk-managed roster drives performance; a fundamentals-focused business sustains demand. Together, they maximize the probability of sustained contention and long-term franchise appreciation.

We stand ready to detail the modeling assumptions and scenario analyses at your convenience.

Respectfully,
Team #2607299

Report on Use of AI Tools

A. AI tools used (disclosure)

During the preparation of this paper, our team used generative AI tools in a limited and assistive manner. The tools and purposes are summarized below.

Tool (Provider)	Use stage	Purpose	Output type / Human check
ChatGPT (OpenAI)	Data collection	Collect/locate public data source URLs relevant to NBA variables used in this paper	URLs only; all links were manually opened and verified; data were downloaded/recorded by the team
ChatGPT (OpenAI)	Writing / typesetting	Assist LaTeX formatting (tables, equations, figure placement) and detect basic grammar/typos	LaTeX snippets; compiled and debugged locally by the team; final formatting decisions made by the team
ChatGPT (OpenAI)	Domain Q&A	Answer NBA-related background questions (definitions, common metrics, context) to support writing	Explanations only; cross-checked against official/statistical sources before adoption

B. How AI outputs were verified (responsibility statement)

We acknowledge that AI-generated content may contain errors or unsupported claims. Therefore:

- For **data source URLs**, we manually opened each link, checked the publisher/host, and validated that the dataset/endpoint matches the described variables before using any numbers.
- For **LaTeX assistance**, we compiled all AI-suggested code locally and corrected any compilation or layout issues; AI outputs were treated as drafts only.
- For **NBA-related explanations**, we verified key facts (metric definitions, rules, and context) using trusted references; any unverified content was discarded.

All modeling choices, computations, results, and conclusions in the paper were produced and confirmed by the team.

C. What AI was *not* used for

AI tools were **not** used to:

- generate final numerical results without human re-computation,
- fabricate datasets or citations,
- write the full solution without team control and revision.

D. Key interaction log (representative records)

Below we list representative prompts/uses (abridged) and how the outputs were adopted. Full chat logs can be provided if required.

1. **Data source discovery (URLs)**: "Find public websites/APIs where NBA team attendance and social following can be obtained; provide URLs and brief descriptions."
   Adoption: Only the URLs were used. We manually verified each source and extracted data ourselves.
2. **LaTeX formatting**: "Convert this table into a LaTeX table with proper alignment and caption; keep within page width."
   Adoption: We used the generated LaTeX as a starting point, then adjusted column widths, captions, and references after compiling.
3. **NBA background Q&A**: "Explain the definition and typical interpretation of eFG, and common caveats when comparing across teams."
   Adoption: We used the explanation only after verifying key statements with reliable references; otherwise we omitted it.

E. Citation and documentation practice

In accordance with COMAP's AI policy, we disclosed AI usage in this section and ensured that all adopted AI-assisted content was verified and revised by the team. :contentReference[oaicite:1]index=1